

A Counterexample for Zero APEPs in Infinite ℓ_1 -Sums

Automated Math Discovery Smoke Test

June 8, 2026

Source and Open Question

This packet concerns Question 8.1 from:

Ramon J. Aliaga, Luis C. Garcia-Lirola, Juan Guerrero-Viu, Matias Raja, Abraham Rueda Zoca, *Almost preserved extreme points*, arXiv:2510.21234.

The source location is page 37 of the paper.

ALMOST PRESERVED EXTREME POINTS

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8. CONCLUDING REMARKS AND OPEN QUESTIONS

In this section we collect some comments and open questions which are derived from our work.

From our study of APEP in the unit ball of ℓ_1 -sums of spaces, the following remains open.

Question 8.1. Let $\{X_i : i \in I\}$ an arbitrary infinite family of Banach spaces and let X be its ℓ_1 -sum. If $0 \in B_X$ is APEP in B_X , must there exist i such that $0 \in X_i$ is APEP in B_{X_i} ?

For the case of ℓ_p -sums, in Proposition 4.9 we have characterised when the norm-one elements are APEP of the unit ball. However, we were unable to describe those APEP whose norm is strictly smaller than 1.

Question 8.1 asks:

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We give a negative answer.

Definitions

Let C be a bounded closed convex subset of a Banach space E . A point $x \in C$ is an almost preserved extreme point, abbreviated APEP, if every relatively weakly open subset W of C containing x contains a non-empty slice of C . For the unit ball, a slice has the form

$$S(B_E, f, \alpha) = \{y \in B_E : f(y) > \sup f(B_E) - \alpha\},$$

where $f \in E^*$ and $\alpha > 0$.

Counterexample

Theorem 1. *There is a countably infinite family of Banach spaces (X_n) such that 0 is an APEP of the unit ball of $X = (\bigoplus_{n=1}^{\infty} X_n)_1$, but 0 is not an APEP of the unit ball of any coordinate X_n .*

Proof. For each $n \geq 1$, let $X_n = H_n = \mathbb{R}^n$ with its Euclidean norm, and put

$$X = \left(\bigoplus_{n=1}^{\infty} H_n \right)_1.$$

Then

$$X^* = \left(\bigoplus_{n=1}^{\infty} H_n \right)_{\infty},$$

where we identify each H_n^* with H_n by the inner product.

First, 0 is not an APEP of B_{H_n} for any fixed n . Indeed, in finite dimension the weak and norm topologies coincide, so

$$W_n = \{h \in B_{H_n} : \|h\| < 1/2\}$$

is a relatively weakly open neighbourhood of 0 in B_{H_n} . No non-empty slice of B_{H_n} is contained in W_n : if $S(B_{H_n}, f, \alpha)$ is non-empty, then either it is all of B_{H_n} in the degenerate case, or it contains a unit vector at which f attains its norm. In both cases it contains a point of norm 1.

It remains to show that 0 is an APEP of B_X . Let W be a relatively weakly open neighbourhood of 0 in B_X . After shrinking W , we may assume that

$$W = \{x \in B_X : |x_j^*(x)| < \varepsilon, \quad j = 1, \dots, m\}$$

for some $x_1^*, \dots, x_m^* \in X^*$ and $\varepsilon > 0$. Write

$$x_j^* = (a_{j,n})_{n \geq 1}, \quad a_{j,n} \in H_n,$$

and let

$$M = \max_{1 \leq j \leq m} \|x_j^*\|.$$

If $M = 0$, any non-empty slice of B_X is contained in W , so assume $M > 0$.

Choose $n > m$. Since $\dim H_n = n$, there is a unit vector $u \in H_n$ orthogonal to the finite set

$$a_{1,n}, a_{2,n}, \dots, a_{m,n}.$$

Define $\varphi \in X^*$ by

$$\varphi((h_k)_{k \geq 1}) = \langle h_n, u \rangle.$$

Choose $\delta > 0$ so small that

$$M(\sqrt{2\delta} + \delta) < \varepsilon.$$

Consider the slice

$$S = \{x = (h_k) \in B_X : \varphi(x) > 1 - \delta\}.$$

This slice is non-empty, since it contains the vector supported only at coordinate n with value u .

We claim that $S \subseteq W$. Let $x = (h_k) \in S$. Since

$$1 - \delta < \langle h_n, u \rangle \leq \|h_n\| \leq \sum_{k=1}^{\infty} \|h_k\| \leq 1,$$

we have

$$\sum_{k \neq n} \|h_k\| < \delta.$$

Also, using $\|h_n\| \leq 1$ and $\langle h_n, u \rangle > 1 - \delta$,

$$\|h_n - u\|^2 = \|h_n\|^2 + 1 - 2\langle h_n, u \rangle \leq 2\delta.$$

For each $j = 1, \dots, m$, orthogonality gives

$$\langle a_{j,n}, h_n \rangle = \langle a_{j,n}, h_n - u \rangle.$$

Therefore

$$\begin{aligned} |x_j^*(x)| &\leq |\langle a_{j,n}, h_n \rangle| + \sum_{k \neq n} |\langle a_{j,k}, h_k \rangle| \\ &\leq M \|h_n - u\| + M \sum_{k \neq n} \|h_k\| \\ &< M(\sqrt{2\delta} + \delta) < \varepsilon. \end{aligned}$$

Thus $x \in W$ for every $x \in S$, so $S \subseteq W$.

Every relatively weakly open neighbourhood of 0 in B_X contains a non-empty slice of B_X . Hence 0 is an APEP of B_X , even though no coordinate has 0 as an APEP. This gives a negative answer to Question 8.1. $\square$

Verification Notes

The argument is purely analytic. The only quantitative estimate is the Hilbert-space fact that if $\langle h, u \rangle > 1 - \delta$ with $\|h\| \leq 1 = \|u\|$, then $\|h - u\| \leq \sqrt{2\delta}$. The script `code/check_slice_estimate.py` performs randomized finite dimensional checks of the orthogonal-vector construction and this bound. It is a sanity check only; the proof above is the certificate.

Limitations and Novelty Check

This packet answers the source question as stated and in its intended infinite-sum setting: the counterexample is a genuine infinite ℓ_1 -sum, not a degenerate one-dimensional edge case. I have not performed a literature search beyond the source paper and the local pipeline corpus, so the novelty status is “candidate” pending human and literature review.

Human Review: Zero APEPs in Infinite ℓ_1 -Sums

Original automatic proof: `solution_packet.pdf`

Checked date: 17 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears to give a correct negative answer to Question 8.1 of the source paper. The construction takes

$$X = \left(\bigoplus_{n=1}^{\infty} \ell_2^n \right)_1.$$

Each coordinate space is finite-dimensional, so 0 is not an APEP of its unit ball: weak and norm topologies coincide, and a sufficiently small norm-neighbourhood of 0 in the ball cannot contain a non-empty slice.

The interesting part is the proof that 0 is nevertheless an APEP of B_X . The argument essentially leverages the finite-dimensional structure of the summands together with the fact that a weak neighbourhood of 0 in B_X is controlled by only finitely many functionals. Given such a neighbourhood, determined by m functionals, one moves to a coordinate $n > m$. In that coordinate, the m corresponding coordinate functionals have a non-trivial common kernel. Thus one can choose a unit vector $u \in \ell_2^n$ invisible to all of them.

The slice is then chosen using the support functional associated to u . Elements in this slice must have almost all of their ℓ_1 -sum norm concentrated in the n -th coordinate. Consequently the other coordinates contribute very little to the finitely many functionals defining the weak neighbourhood. In the n -th coordinate, the chosen vector u lies in the common kernel, and the Hilbert-space estimate

$$\langle h, u \rangle > 1 - \delta, \quad \|h\| \leq 1 = \|u\| \quad \implies \quad \|h - u\| \leq \sqrt{2\delta}$$

shows that the n -th coordinate is close to u . Hence the contribution from the n -th coordinate is also small. This proves that the slice lies inside the original weak neighbourhood.

Extension to Other Finite-Dimensional Blocks

The proof is not special to Hilbert spaces in its overall structure. The same argument should work with

$$X_n = \ell_p^n$$

for every fixed $1 < p < \infty$. The only replacement needed is for the Hilbert-space estimate above. In ℓ_p^n , choose

$$u \in S_{\ell_p^n} \cap \bigcap_{j=1}^m \ker a_{j,n}$$

and then choose a norming functional $u^* \in S_{(\ell_p^n)^*}$ with $u^*(u) = 1$. Since ℓ_p^n is strictly convex for $1 < p < \infty$, the slice

$$\{h \in B_{\ell_p^n} : u^*(h) > 1 - \delta\}$$

shrinks to u as $\delta \downarrow 0$. In finite dimension this can be justified by compactness. Therefore the same estimates go through after replacing the explicit Hilbert bound by this qualitative slice-shrinking fact.

For $p = 1$, the same argument does not apply, since the unit ball has flat faces and a support functional need not expose a single vector. Thus the natural extension of the proof is to the range $1 < p < \infty$.

Additional Note: Infinite-Dimensional Summands

If one wants the coordinate spaces themselves to be infinite-dimensional, the same construction should admit a straightforward modification. Take

$$X_n = \ell_2^n \oplus_1 c_0$$

and

$$X = \left(\bigoplus_{n=1}^{\infty} X_n \right)_1.$$

The ℓ_2^n -part still provides the same finite-dimensional escape direction. Given a weak neighbourhood of 0 in B_X determined by finitely many functionals, choose n larger than the number of constraints and choose

$$u \in S_{\ell_2^n}$$

orthogonal to the ℓ_2^n -coordinate parts of those functionals. Define the slice using only the ℓ_2^n -component,

$$\varphi((h_k, y_k)_k) = \langle h_n, u \rangle.$$

As before, the condition $\varphi(x) > 1 - \delta$ forces almost all of the ℓ_1 -sum mass to lie in the n -th ℓ_2^n -component. It also forces the c_0 -part inside the n -th coordinate, and all other coordinates, to have small total norm. The same Hilbert-space estimate then puts the slice inside the prescribed weak neighbourhood.

The c_0 -summand is only used to make each X_n infinite-dimensional while keeping 0 visibly non-APEP in each coordinate. One can witness this with a weak neighbourhood in B_{X_n} which controls both components, for instance

$$W_n = \{(h, y) \in B_{\ell_2^n \oplus_1 c_0} : \|h\|_2 < 1/2, |y_1| < 1/2\}.$$

No non-empty slice of B_{X_n} is contained in W_n : if the ℓ_2^n -part of the defining functional dominates, the slice meets a point of the form $(h, 0)$ with $\|h\|_2 = 1$; if the c_0 -part dominates, the slice meets a point of the form $(0, y)$ with $|y_1| = 1$, using the standard ℓ_1 -description of c_0^* . Thus 0 is not an APEP of B_{X_n} for any n , while the ℓ_1 -sum still has 0 as an APEP.

Review Outcome

Archive this packet as an apparently correct counterexample. The proof's central mechanism is sound: every coordinate fails APEP at 0 because it is finite-dimensional, while the infinite ℓ_1 -sum allows one to escape any finite list of weak constraints by moving to a sufficiently high-dimensional coordinate and choosing a direction annihilated by those constraints. The same construction should also work with ℓ_p^n blocks for $1 < p < \infty$.